Fpt University

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CAPSTONE PROJECT REPORT

Report 2 – Project Management Plan

– FPT University, October 2026 –

Table of Contents

I. Record of Changes....3
II. Project Management Plan....4
1. Overview....4
1.1 Scope & Estimation....4
1.2 Project Objectives....4
1.3 Project Risks....4
2. Management Approach....5
2.1 Project Process....5
2.2 Quality Management....5
2.3 Training Plan....5
3. Project Deliverables....5
4. Responsibility Assignments....5
5. Project Communications....6
6. Configuration Management....6
6.1 Document Management....6
6.2 Source Code Management....6
6.3 Tools & Infrastructures....6

I. Record of Changes

Date	A* M, D	In charge	Change Description
26/10/2026	M	Lê Nguyễn Quang Đức	Populate VISTA Review 2 content on original university template; code verification pending

*A - Added M - Modified D - Deleted

II. Project Management Plan

1. Overview

1.1 Scope & Estimation

Workstreams: WF-01 school/robot allocation; WF-02 classroom learning and submission; WF-03 schematic experiment and runtime integration; WF-04 assessment/AI metadata. Effort estimates remain team estimates pending sign-off, not timesheets.

Effort and complexity are estimated by workflow and technical risk; the official Project Tracking WBS is maintained in its existing WBS, Issues, Defects and Q&A tabs.

#	WBS Item	Complexity	Est. Effort (man-days)
1	Access/organization	Medium (planning estimate)	Estimate pending team validation
2	Learning and assignment	Medium (planning estimate)	Estimate pending team validation
3	Experiment/runtime	Medium (planning estimate)	Estimate pending team validation
4	Assessment/AI metadata	Medium (planning estimate)	Estimate pending team validation
5	SRS and SDD	Medium (planning estimate)	Estimate pending team validation
6	CI/test evidence	Medium (planning estimate)	Estimate pending team validation
		Medium (planning estimate)	Estimate pending team validation

1.2 Project Objectives

Review 2: trace Registration v1.2 to SRS/SDD, code/schema and test plan. Defense 1.1: demonstrate four main workflows with main and exception paths on declared staging, using documented targets rather than fabricated passes.

Objectives: demonstrate a reproducible Iteration 1, maintain consistency among Registration, SRS, SDD, migrations, tests and CI. Targets for Defense 1.1 are recorded in BM01 Appendix A and become binding only after Review 2.

Quality

#	Testing Stage	Test Coverage	No. of Defects	% of Defect	Notes
1	Reviewing				
2	Unit Test				
3	Integration Test				
4	System Test				
5	Acceptance Test				

Quality: authorization denial must be tested; physical safety must fail closed; no invented experiment execution success. Track actual test counts separately.

Milestone timeliness: to be measured against supervisor-approved schedule; no achieved percentage asserted.

Allocated effort: owner estimates to be approved; no fabricated actual effort.

1.3 Project Risks

Major risks: mismatched ERD cardinality/state transitions, physical robot integration gaps, insufficient CI/test evidence and multi-organization authorization leakage.

#	Risk Description	Impact	Possibility	Response Plans
1	Physical runtime integration unverified	High	High	Separate schematic request from hardware execution and collect tested E2E evidence

2. Management Approach

Iterative development across four owner-led workflows. Design revisions and code changes use PR review; Registration scope is frozen only after Review 2 approval.

2.1 Project Process

Iterative delivery, feature branches/PR review, synchronized FR-to-code-to-test changes, review-2 Git tag for immutable self-review baseline.

Process: Requirements → SRS/SDD and diagram review → feature branch implementation → unit/integration tests → PR/CI → staging demo → issue tracking.

2.2 Quality Management

Review each feature against registration, actor, scope guard, state path, database constraints and test cases. Never mark a test Passed without observed run evidence.

- Prevent defects with validation, least-privilege scope tests, immutable program versions and verified migrations.
- Review PR changes and diagram/state alignment prior to merge.
- Unit tests: NestJS Jest, frontend Vitest, Python runtime tests where available.
- Integration tests: API/DB authorization and attempt–program–submission invariants.

- System tests: the four workflow main and exception demonstrations using seeded staging data.

2.3 Training Plan

Learning topics: hardware safety, Prisma migrations, Supabase membership authorization, CI workflows and test traceability; plan training as gaps are confirmed.

Training Area	Participants	When, Duration	Waiver Criteria
Pending team verification			Mandatory
Git, Github			Mandatory

3. Project Deliverables

Registration v1.2 + BM03, reports 1–5, BM01/BM02, 4 workflow sources and C4/ERD/state diagrams, seed/database script, CI/test evidence and project tracking. BM04 due after Review 2.

Deliverables: registration with BM03, Reports 1–5, Project Tracking, PlantUML/Mermaid diagrams, SQL migrations, code and test evidence, BM02 and BM01 Appendix A.

#	Deliverable	Due Date	Notes
1	...	Pending team verification	…

4. Responsibility Assignments

Decisions and workflow ownership: Đức WF-01/auth/organization; Đạt WF-02/curriculum/submission; Tuấn WF-03/robot runtime; Lợi WF-04/AI metadata/assessment. Real delivery and PR assignments must be verified.

Ownership: Đức WF-01 and integration; Đạt WF-02 and course delivery; Tuấn WF-03 and robot runtime; Lợi WF-04, assessment and ERD.

D~Do; R~Review; S~Support; I~Informed; <blank>- Omitted

Responsibility	Lê Nguyễn Quang	Trần Tiến Đạt	Vũ Anh Tuấn	Phạm Viết Lợi	ThuyVT04278
WF-01	D	R	R	R	R
WF-02	R	D	R	R	I
WF-03	R	R	D	R	R
WF-04	R	R	R	D	R

SRS/SDD	R	R	R	D	

5. Project Communications

Communication Item	Who/ Target	Purpose	When, Frequency	Type, Tool, Method(s)

Weekly meetings; all members attend Review 2, prepare code explanations and ownership evidence.

6. Configuration Management

6.1 Document Management

Keep Word/Excel on official university templates, diagrams sourced from user-approved PDF, editable source diagram versions in repo; use SEP490_FA26SE183_R2_ filenames and seven-folder submission structure.

Version Word/PDF documents with project code FA26SE183; editable diagrams must correspond to repository source. Maintain explicit Draft versus Approved status.

6.2 Source Code Management

Source repository https://github.com/SKYNET-Thesis/se (main); do not tag review-2 until CI, docs and schema have been checked. BM02 must analyze exact tagged commit.

Repository: https://github.com/SKYNET-Thesis/se. Work in feature branches, review via PR; tag review-2 only after code/documents are reconciled. Never rewrite tag after official self-review.

Technology: React 18/Vite, NestJS 11, Prisma/PostgreSQL, Supabase Auth, Python Robot Runtime, Docker, GitHub Actions. AWS staging is planned and must not be reported as deployed without evidence.

Category	Tools / Infrastructure
Technology	Pending team verification
Database	MySQL, FireBase
IDEs/Editors	Visual Studio Code, Eclipse
Diagramming	StarUML, DrawIO
Documentation	Ms Office, Google Docs/Sheets/Slides
Version Control	GitLab (Source Codes), Google Drive (Documents)
Deployment server	Amazon Web Service
Project management	ProjectLibre (Schedule), GitLab (Tasks, Defects)

Review 2 design verification milestone: validate 22-page diagram source (3 C4, 4 Activity, 4 State Machines, 1 Conceptual ERD, 4 Logical ERDs, 6 Physical ERDs), align Report 3/4/5 references, then review DB constraints and evidence prior to review-2 tag.